

A FOUR-VERTEX COUNTEREXAMPLE TO DEVRIENDT'S CONJECTURE 7.6

ABSTRACT. Devriendt conjectured that, for a normalized weighted graph, submodularity of its resistance capacity implies nonnegative resistance curvature. We give a normalized weighting of K_4 for which all 24 elementary submodularity inequalities are strict, while one curvature coordinate equals $-1/126$. This disproves Conjecture 7.6. The example has minimum possible order and persists under small perturbations. It does not refute the graph-level existential statement in Conjecture 7.7.

Let $G = (V, E)$ be a finite connected simple graph with $|V| \geq 2$ and positive edge conductances c . Let L be its weighted Laplacian and $L^\dagger$ its Moore–Penrose pseudoinverse. With e_u denoting the standard basis vector at u , define

$$\omega_{uv} = (e_u - e_v)^\top L^\dagger (e_u - e_v), \quad \Omega = (\omega_{uv})_{u,v \in V}, \quad K = \Omega^{-1}.$$

Following [1, Definitions 3.1, 5.1, and 7.1], the weighting is *normalized* when $\mathbf{1}^\top K \mathbf{1} = 1$. For $uv \in E$, put $r_{uv} = c_{uv} \omega_{uv}$, and define the resistance curvature by

$$p_v(c) = 1 - \frac{1}{2} \sum_{u: uv \in E} r_{uv}.$$

For normalized c , the resistance capacity is the set function $\tau_\bullet(c)$ given by

$$\tau_\emptyset = 0, \quad \tau_{\{v\}} = \frac{1}{2}, \quad \tau_U = \frac{1}{2} + \frac{1}{2} \left(\mathbf{1}_U^\top \Omega[U]^{-1} \mathbf{1}_U \right)^{-1} \quad (|U| \geq 2),$$

where $\Omega[U]$ is the principal submatrix indexed by U and $\mathbf{1}_U$ is the all-ones vector in $\mathbb{R}^U$. Devriendt proved that $p(c) \geq 0$ implies submodularity of $\tau_\bullet(c)$ and conjectured the converse [1, Theorem 7.5 and Conjecture 7.6].

Theorem 1. *Let $G = K_4$ on $V = \{0, 1, 2, 3\}$, order the edges as 01, 02, 03, 12, 13, 23, and set*

$$c = \frac{3485}{15876} (1, 1, 2, 1, 4, 4).$$

Then c is normalized, $\tau_\bullet(c)$ is submodular, and

$$p(c) = \left(\frac{47}{126}, \frac{20}{63}, \frac{20}{63}, -\frac{1}{126} \right).$$

Moreover, every elementary submodularity slack is positive, with minimum 1123/69700. Hence Conjecture 7.6 of [1] is false.

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Proof. Write $w = (1, 1, 2, 1, 4, 4)$. Its Laplacian is

$$L(w) = \begin{pmatrix} 4 & -1 & -1 & -2 \\ -1 & 6 & -1 & -4 \\ -1 & -1 & 6 & -4 \\ -2 & -4 & -4 & 10 \end{pmatrix}.$$

Using

$$L^\dagger = \left(L + \frac{1}{4}\mathbf{1}\mathbf{1}^\top\right)^{-1} - \frac{1}{4}\mathbf{1}\mathbf{1}^\top,$$

one obtains

$$(1) \quad \Omega(w) = \frac{1}{126} \begin{pmatrix} 0 & 44 & 44 & 35 \\ 44 & 0 & 36 & 23 \\ 44 & 36 & 0 & 23 \\ 35 & 23 & 23 & 0 \end{pmatrix}, \quad \mathbf{1}^\top \Omega(w)^{-1} \mathbf{1} = \frac{15876}{3485}.$$

Since $\Omega(\lambda w) = \lambda^{-1}\Omega(w)$, the stated choice $c = (3485/15876)w$ satisfies $\mathbf{1}^\top \Omega(c)^{-1} \mathbf{1} = 1$.

The relative resistances, which are invariant under common rescaling of the conductances, are

$$(r_{01}, r_{02}, r_{03}, r_{12}, r_{13}, r_{23}) = \left(\frac{22}{63}, \frac{22}{63}, \frac{5}{9}, \frac{2}{7}, \frac{46}{63}, \frac{46}{63}\right).$$

Substitution in the definition of p gives the displayed curvature vector.

For distinct $i, j \notin S$, define

$$\Delta_{ij}(S) = \tau_{S \cup \{i\}} + \tau_{S \cup \{j\}} - \tau_S - \tau_{S \cup \{i, j\}}.$$

Submodularity is equivalent to the nonnegativity of the $\binom{4}{2}2^2 = 24$ quantities $\Delta_{ij}(S)$. The weighting is invariant under $1 \leftrightarrow 2$. Since $\Omega(c) = (15876/3485)\Omega(w)$, exact inversion of the principal submatrices in (1) gives the following orbit representatives. In the table, strings such as 23 denote the corresponding vertex sets.

ij	S	$\Delta_{ij}(S)$
01	$\emptyset, 2, 3, 23$	$\frac{713}{6970}, \frac{3888}{17425}, \frac{2107}{20910}, \frac{2389}{41820}$
03	$\emptyset, 1, 12$	$\frac{128}{697}, \frac{112}{615}, \frac{1123}{69700}$
12	$\emptyset, 0, 3, 03$	$\frac{1217}{6970}, \frac{5148}{17425}, \frac{207}{2788}, \frac{64}{2091}$
13	$\emptyset, 0, 2, 02$	$\frac{1018}{3485}, \frac{3038}{10455}, \frac{2673}{13940}, \frac{1346}{52275}$

The involution $1 \leftrightarrow 2$ supplies the four cases for 02, the four cases for 23, and $\Delta_{03}(\{2\}) = \Delta_{03}(\{1\})$, accounting for all 24 inequalities. Every displayed value is positive, and

$$\min_{\substack{i < j \\ S \subseteq V \setminus \{i, j\}}} \Delta_{ij}(S) = \Delta_{03}(\{1, 2\}) = \frac{1123}{69700} > 0.$$

Thus $\tau_\bullet(c)$ is submodular, whereas $p_3(c) < 0$. □

Remark 2 (Minimality and stability). The connected simple graphs on fewer than four vertices are K_2 , P_3 , and K_3 . On a weighted tree, each edge has relative resistance 1, so $p_v = 1 - \deg(v)/2 \geq 0$ on K_2 and P_3 . For a weighted triangle, let e, f be the edges incident with v , let g be the opposite edge, and put $x_h = c_h^{-1}$ and $X = x_e + x_f + x_g$. The series-parallel law gives $r_h = 1 - x_h/X$, and hence

$$p_v = 1 - \frac{1}{2}(r_e + r_f) = \frac{x_e + x_f}{2X} > 0.$$

Thus order four is minimal. Since curvature and the finitely many elementary slacks depend continuously on positive conductances, the strict inequalities $p_3(c) < 0$ and $\min \Delta_{ij}(S) > 0$ persist under sufficiently small perturbations, followed by normalization.

Remark 3 (Scope). Conjecture 7.6 is pointwise in the normalized conductance vector. The displayed Conjecture 7.7 of [1] is existential over conductance vectors, so the two formulations have different quantifier structures. The example above directly refutes Conjecture 7.6, but not the graph-level statement in Conjecture 7.7: uniform conductances on K_4 , after normalization, have curvature $(1/4, 1/4, 1/4, 1/4)$, and hence K_4 is resistance nonnegative.

REFERENCES

- [1] K. Devriendt, *Graphs with nonnegative resistance curvature*, Ann. Comb. **30** (2026), 415–438. doi:10.1007/s00026-025-00774-x.